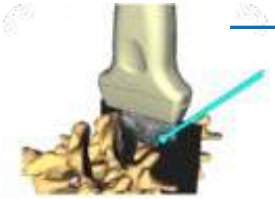
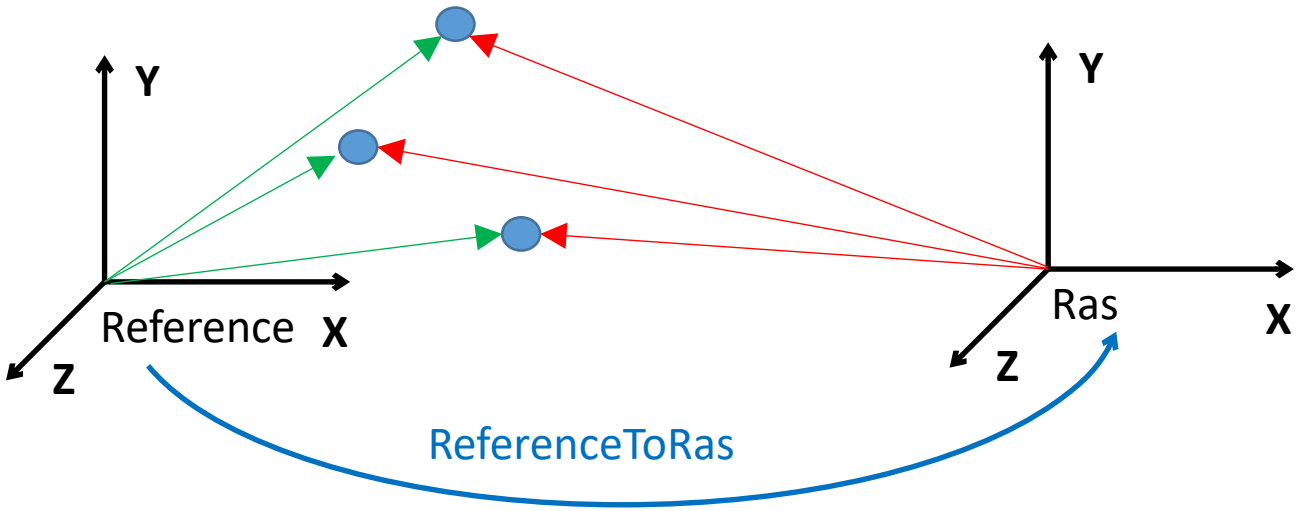
Landmark registration



SlicerIGT Tutorial Series

Landmark registration

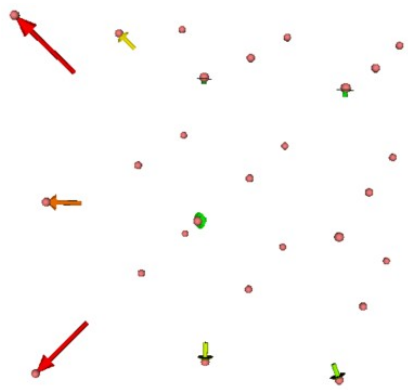
- Computes transformation between point pairs
- Points = landmarks = fiducials
- Computation is fast, non-iterative, optimal result
- Sensitive to outliers!
- **vtkLandmarkTransform** based on [Horn 1987]



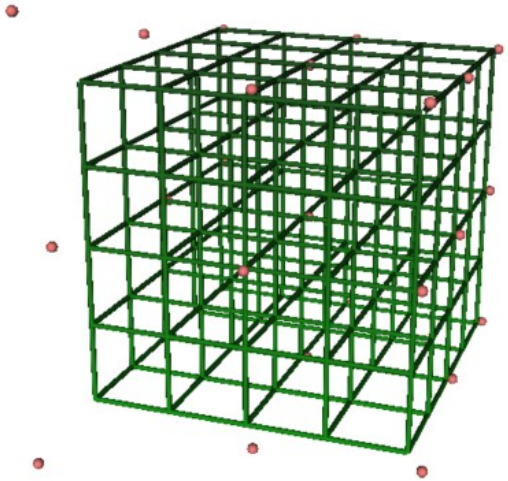
Result transform type

- Rigid
  - Translation + Rotation
  - Commonly used, because objects don't change size
- Similarity
  - Translation + Rotation + Scaling
  - Used when changing units, e.g. pixels to millimeters
- Warping
  - Splines, etc.
  - Used to model soft tissue deformations

Fiducial displacement:

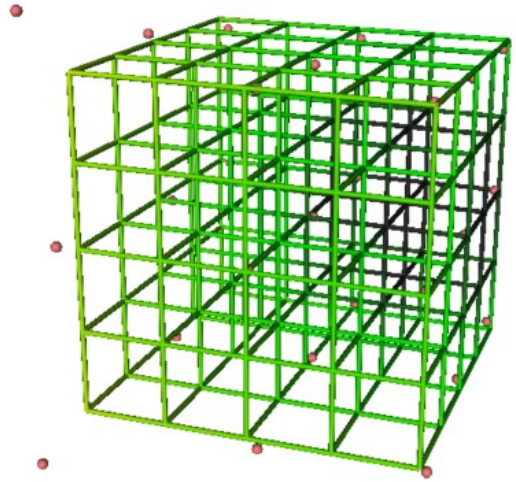


Rigid



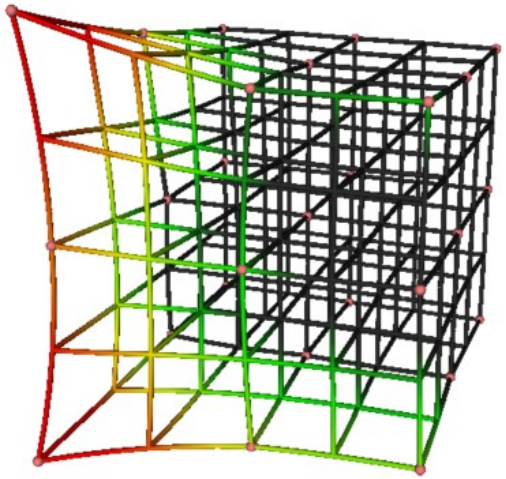
RMSE = 4.8

Similarity



RMSE = 4.2

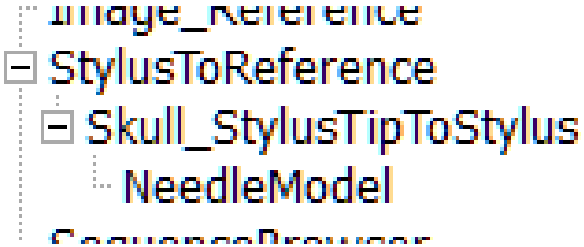
Warping



RMSE = 0.0

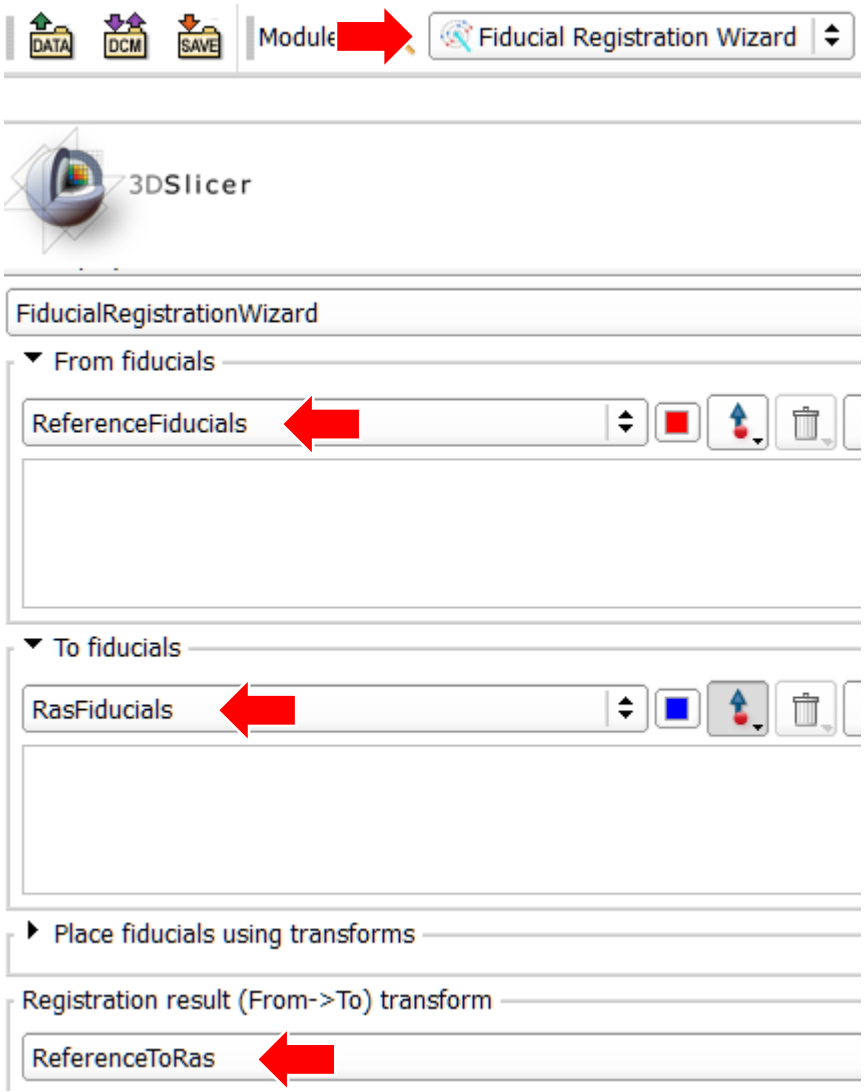
Load example data

- Load scene:
  SlicerIGT-Data/Skull_GroundTruthLandmarks.mrb
- Load additional transform:
  SlicerIGT-Data/Skull_StylusTipToStylus.h5
- Create needle model
- Set up stylus transform hierarchy in **Data** module:



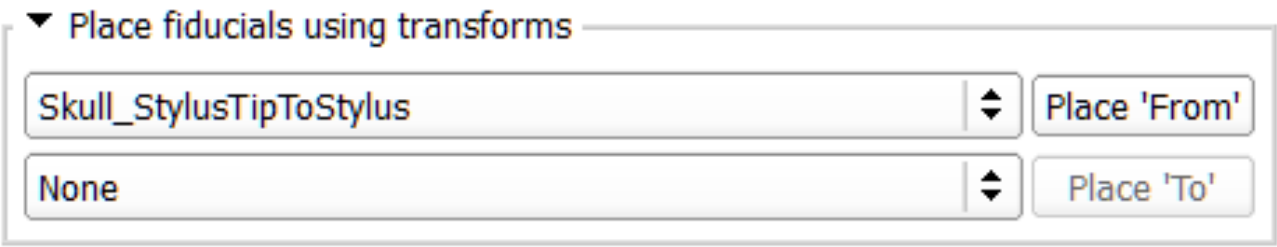
Create input and output nodes

- Select **IGT / Fiducial Registration Wizard** module
- Create *From* fiducials and name the list "**ReferenceFiducials**"
- Create *To* fiducials and name the list "**RasFiducials**"
- Create *result transform* as "**ReferenceToRas**"



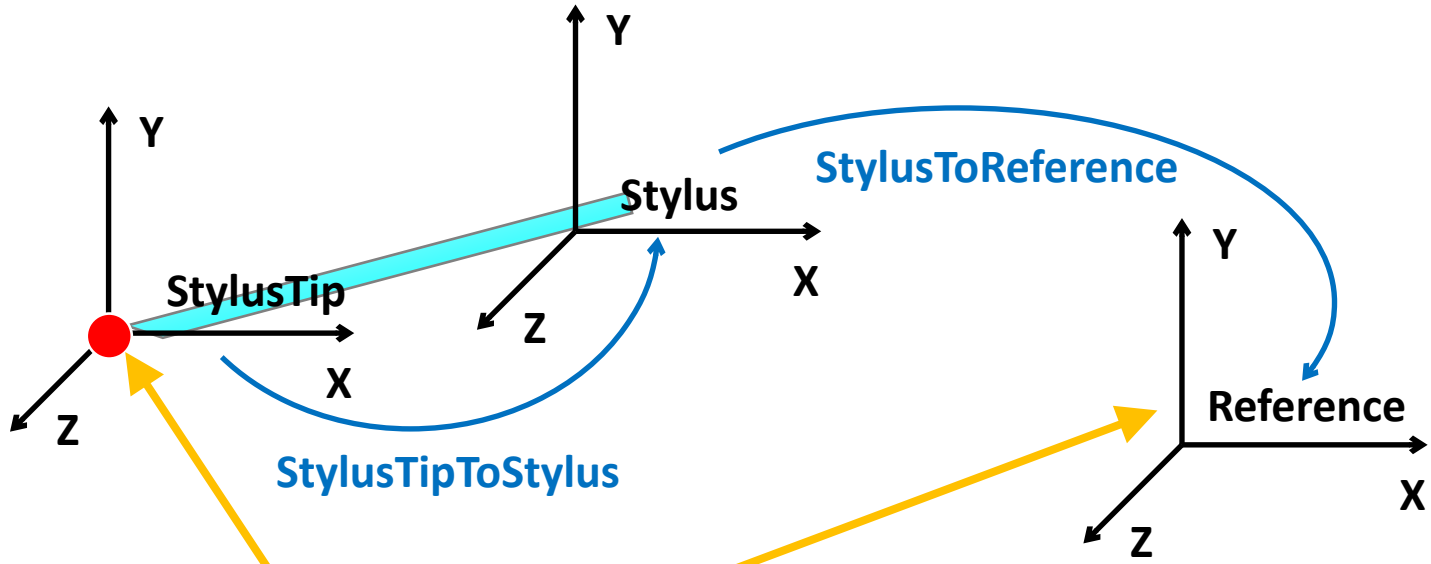
Set up stylus as input

- Expand **Place fiducials using transforms** group
- Select **Skull_StylusTipToStylus** to place *From* fiducials



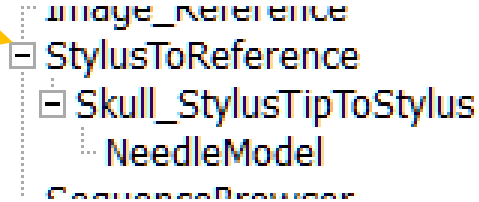
- When pressing **Place 'From'** or **Place 'To'** buttons, fiducials are created at the *origin* (0,0,0) of the *From* (first) coordinate system of the selected transform, transformed by the transform hierarchy.
- See explanation on next slide

How *Place fiducials...* work



Place 'From' adds **this** point in **this** coordinate system to **this** list

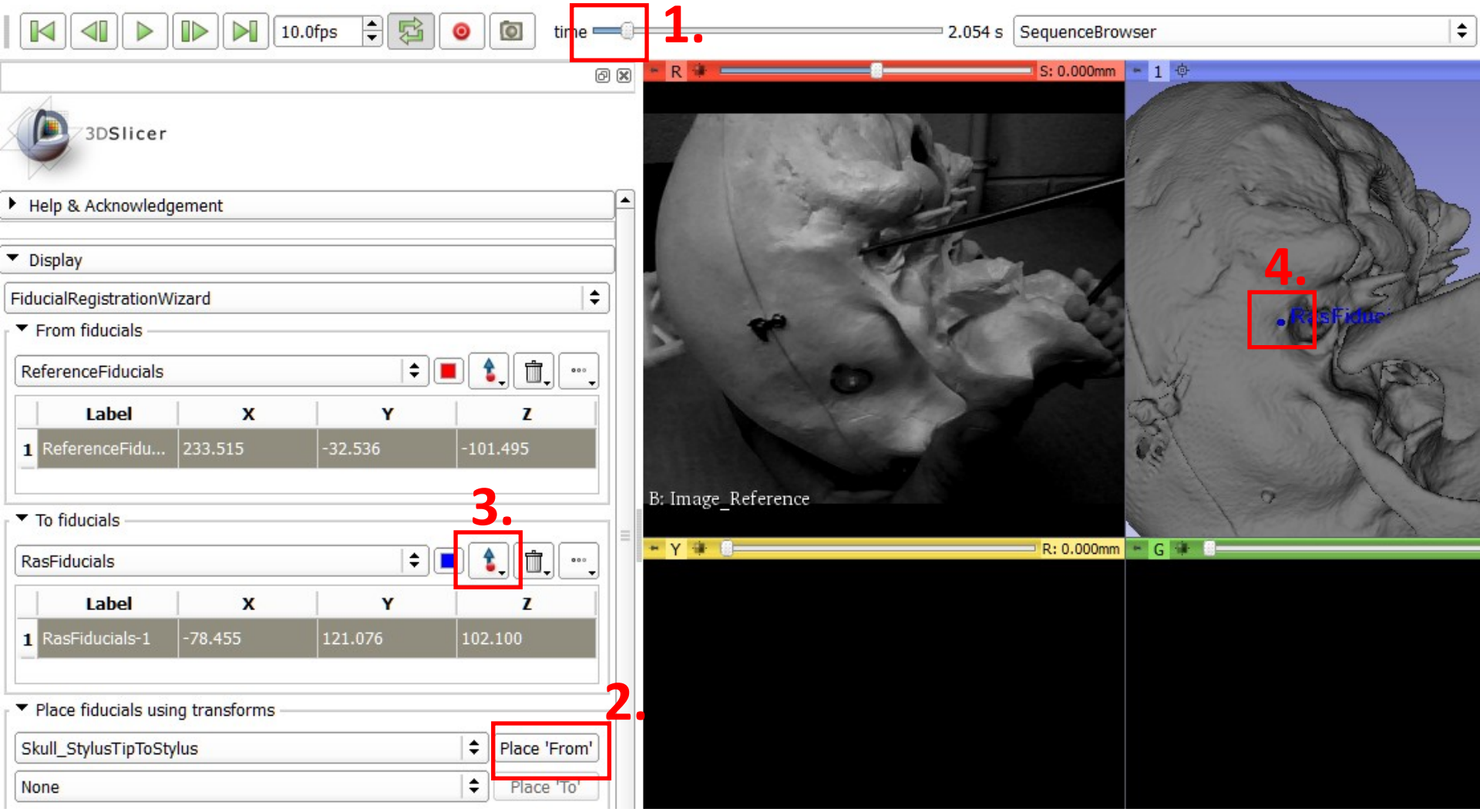
because of this selection and this transform hierarchy



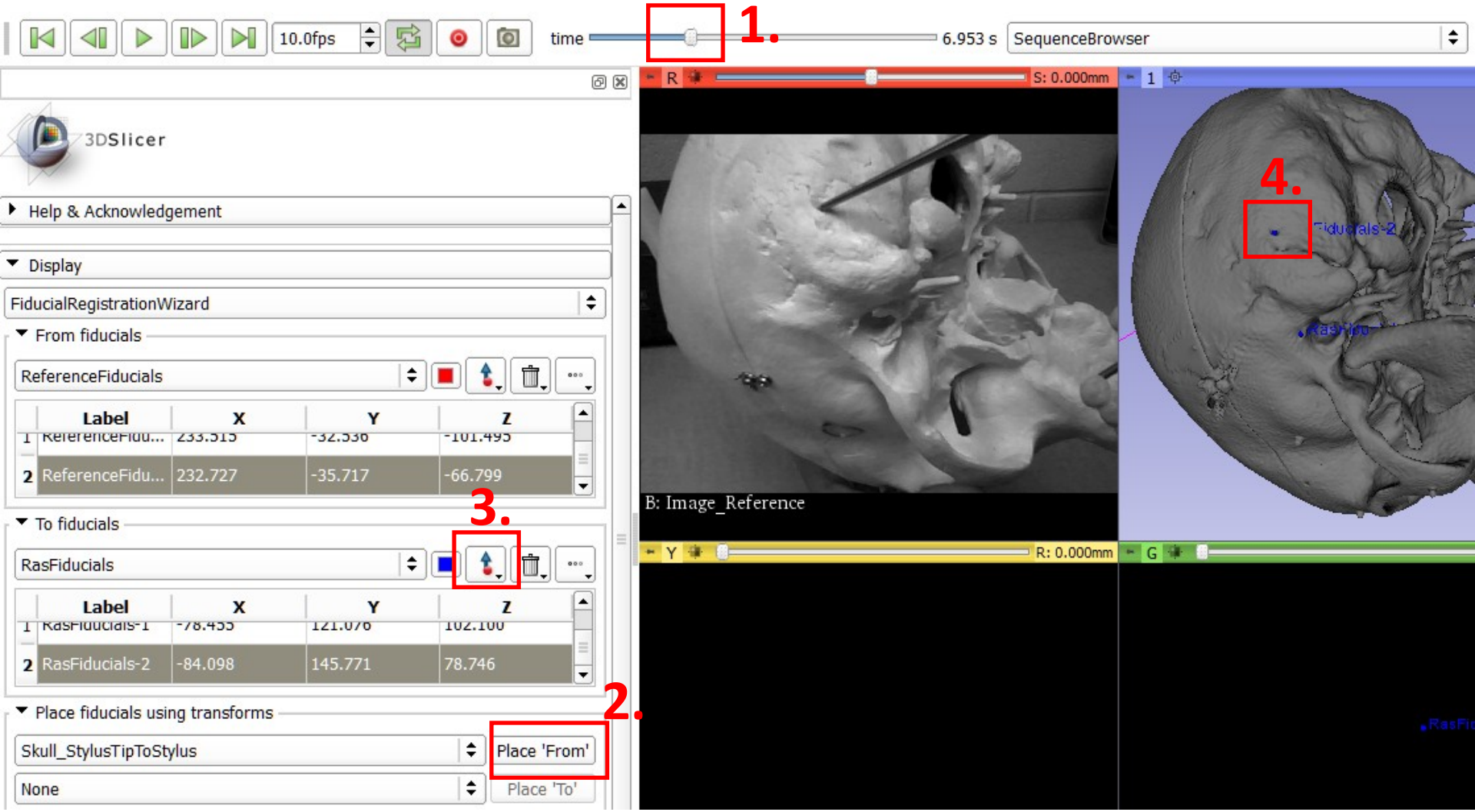
Populate the *'From'* and *'To'* lists

- Select **SequenceBrowser** node in the Sequence toolbar
- Drag the **time** slider to points where you see the stylus placed in skull landmarks in the video
- Rotate the 3D view so you see the same landmark point in the surface model
- Press the **Place 'From'** button to add a *From* fiducial
- Press the fiducial selector button in the To list and click on surface model to place a *To* fiducial
- See next slide for illustration

Adding first point pair



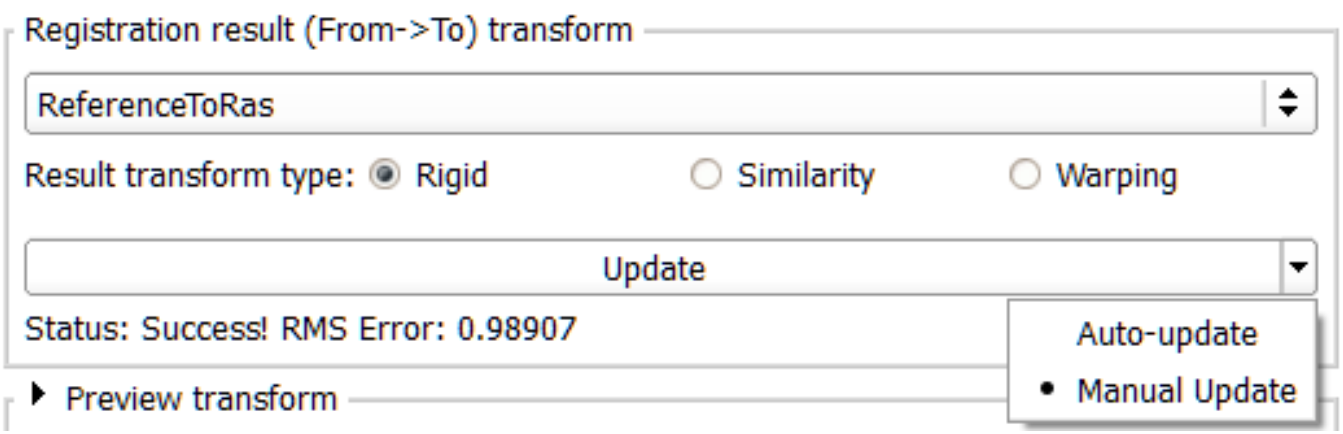
Adding second point pair



And so on for many point pairs…

… but how many?

- At least 3 point pairs are needed for registration
- More point will probably increase accuracy AND slightly increase the RMSE (root mean square error)
- Accuracy will increase because random errors cancel each other out
- RMSE will increase because of less overfitting
- RMSE is an <u>unknown</u> mix of precision and accuracy!

When all point pairs are added

- It is a good idea to switch to **Manual update** from **Auto-update** to prevent errors when accidentally moving fiducials later in the scene
- Save the **ReferenceToRas** transform, or save the entire scene, in case you want to check the result later